

Counting domino tilings of a $3 \times 2n$ rectangle

a worked case of research mode: guess, test, prove twice

Abstract

Let $a(n)$ be the number of ways to tile a $3 \times 2n$ board with dominoes. The problem has no answer key in this case file, so every claim carries its own evidence. Counts from three independent methods agree up to $n = 16$. A recurrence guessed from 13 terms survives four held-out terms and an exhaustive check to $n = 60$, and is then proved in two independent ways: by a column decomposition and by an order bound plus a finite check. The result is classical; the point of the case is the workflow and the honest labelling of each step.

1 Result

Theorem 1. $a(0) = 1$, $a(1) = 3$ and $a(n) = 4a(n-1) - a(n-2)$ for $n \geq 2$. Equivalently

$$a(n) = \frac{3 + \sqrt{3}}{6} (2 + \sqrt{3})^n + \frac{3 - \sqrt{3}}{6} (2 - \sqrt{3})^n, \quad \frac{a(n+1)}{a(n)} \rightarrow 2 + \sqrt{3}.$$

The sequence 1, 3, 11, 41, 153, ... is OEIS A001835 shifted by one index; the case did not search the literature further, and makes no claim of novelty.

2 How it was found

Three independent counts. A backtracking program fills the first free cell of a board, a transfer matrix over the $2^3 = 8$ column profiles counts tilings by matrix powers, and the coupled recurrences of Section 3 count them by induction. Backtracking and the transfer matrix agree for $n \leq 5$ (the six boards 3×0 to 3×10); the transfer matrix and the recurrences agree for $n \leq 16$. Boards with one corner removed agree between backtracking and the recurrence for $1 \leq m \leq 8$.

Guess, then test. From $a(0), \dots, a(12)$ an exact search finds a recurrence of order 2 with coefficients $(4, -1)$: eleven equations, two unknowns. No polynomial of degree ≤ 8 fits (the growth is exponential, Figure 1). Held out from the fit, $a(13), \dots, a(16)$ all satisfy the recurrence, and an exhaustive search over $2 \leq n \leq 60$ finds no counterexample.

3 Proof A: decomposing the board

Let A_m be the number of tilings of the $3 \times m$ board and B_m the number for the same board with one corner cell removed; set $A_0 = 1$, $A_1 = 0$, $B_0 = 0$, $B_1 = 1$.

$A_m = A_{m-2} + 2B_{m-1}$. Look at the three cells of the leftmost column. Either all three are covered by horizontal dominoes, which leaves a $3 \times (m-2)$ board; or a vertical domino covers the top two cells, which forces a horizontal domino on the bottom cell and leaves a $3 \times (m-1)$ board with its bottom-left corner removed; or the mirror image of that case. These cases are exhaustive and disjoint.

n	$a(n)$	$4a(n-1) - a(n-2)$
0	1	
1	3	
2	11	11
3	41	41
4	153	153
5	571	571
6	2 131	2 131
7	7 953	7 953
8	29 681	29 681
9	110 771	110 771
10	413 403	413 403

Table 1. The first terms and the guessed recurrence.

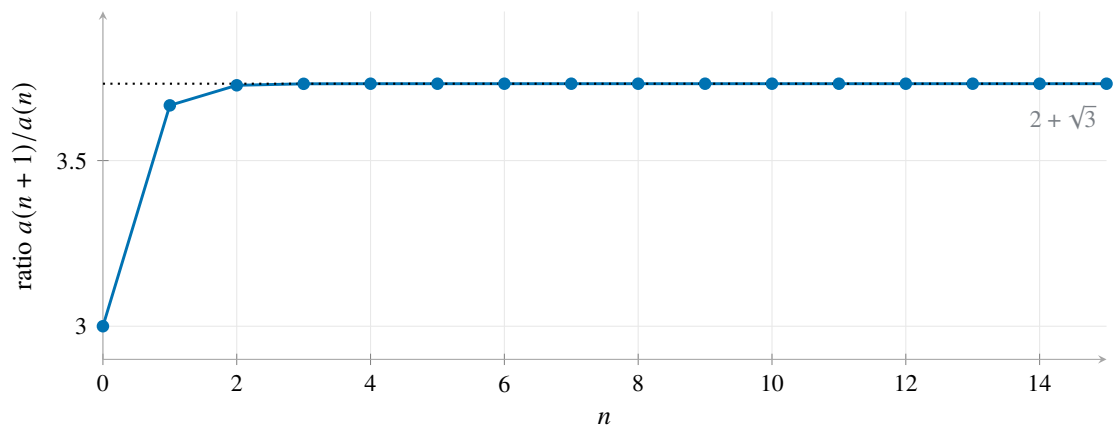


Figure 1. The ratio of consecutive terms against the limit $2 + \sqrt{3} \approx 3.7321$, the larger root of $x^2 - 4x + 1$.

$B_m = A_{m-1} + B_{m-2}$ (corner cell $(3, 1)$ removed). The two remaining cells of the first column are covered by one vertical domino, leaving a full $3 \times (m - 1)$ board; or by two horizontal dominoes, which forces a horizontal domino on the third cell of the second column and leaves a $3 \times (m - 2)$ board with a corner removed.

Eliminating B : $2B_{m-1} = A_m - A_{m-2}$ and $B_{m-1} = A_{m-2} + B_{m-3}$ give $A_m = 4A_{m-2} - A_{m-4}$ for $m \geq 4$. With $a(n) = A_{2n}$ this is Theorem 1. The closed form follows from the roots $2 \pm \sqrt{3}$ of $x^2 - 4x + 1$ and the values $a(0)$, $a(1)$, and the limit of the ratio from $|2 - \sqrt{3}| < 1$. $\square$

4 Proof B: an order bound and a finite check

The number of tilings of $3 \times m$ is $(T^m)_{00}$ for the 8×8 column-profile matrix T , so $a(n) = (T^{2n})_{00}$ is annihilated by a monic polynomial $P(E)$ of degree at most 8 in the shift E (Cayley–Hamilton for T^2). Let $Q(E) = E^2 - 4E + 1$ and $d = Q(E)a$. Polynomials in E commute, so $P(E)d = Q(E)P(E)a = 0$: the difference d obeys the same monic recurrence of order ≤ 8 , and a sequence like that vanishes once its first 8 terms do. Eight consecutive equations, $a(0), \dots, a(9)$, hold; hence $d \equiv 0$. This proof does not use Proof A and is the template implemented by `check_recurrence` with `order_bound=8`.

5 Confidence record

Statement	Evidence	Level
$a(n)$ for $n \leq 5$	backtracking, transfer matrix, recurrences agree	checked to $n = 5$, three methods
$a(n)$ for $n \leq 16$	transfer matrix and recurrences agree	checked to $n = 16$, two methods
$a(n) = 4a(n-1) - a(n-2)$ for all n	Proof A; Proof B; held-out terms; exhaustive to $n = 60$	proved (two independent proofs)
closed form and growth rate	algebra from the recurrence	proved
no polynomial formula	degree ≤ 8 search; exponential growth	proved by growth
novelty	OEIS A001835 lists the sequence	not claimed

Table 2. What each statement rests on, using the confidence ladder of the research-mode guide.

6 Routes tried

Three routes were recorded with the route-graph tool: (i) guess a linear recurrence from data and test it on held-out terms, which survived and became Proof B; (ii) guess a polynomial formula, which was killed because the growth is exponential; (iii) prove the recurrence from the column decomposition, which was chosen as the main proof and cross-checked against backtracking. The partial result “eight consecutive equations prove the recurrence” was kept with the status *proved by finite check*. The lesson written to the project memory reads: when a count looks like a linear recurrence, guess it from data, hold terms out, then prove it twice, from a local decomposition and from an order bound plus a finite check.

7 Limits

A case this small shows the workflow, not research strength. The theorem is classical. The searches support the guess; the proofs rest on the case analysis of Section 3 and on the order bound of Section 4, both short enough to check by hand. Nothing here is checked by a proof assistant.

Reproduce. `uv run --locked python demos/domino-research/reproduce/explore.py` recomputes every number in a few seconds; `build_note.py` typesets this note from the archived record.